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School Name: University of Lancashire

Assessment title: WellSpan-Health Monitoring System – Group Project

Course Title: Electronic Engineering
Module Title: Electronic Systems Applications
Module Code: EL2205
Year of Study: 2026
Student Number: Mo Yan: [redacted]; Lowlife555: [redacted]

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Health Monitoring System – Group Project

By

Mo Yan and Lowlife555

This paper is the major assignment for the EL2205 course

Electrical Engineering
University of Lancashire, UK

March, 2026

University of Lancashire

Comment

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Abstract

This project aims at the home rehabilitation monitoring needs for patients after surgery, chronic diseases, and sports injury recovery period. We developed a wearable auxiliary monitoring system based on the STM32L476RG microcontroller. The system achieves five core monitoring functions: multi-parameter vital signs monitoring (heart rate and SpO2), rehabilitation environment monitoring (temperature and humidity), training body balance monitoring (level meter), hand strength exercise (game), and fall abnormal warning (gravity accelerometer). Also, the system is equipped with the Bluetooth module to transmit measured data to the computer web page, achieving the rehabilitation data visualization.

The project strictly followed assignment requirements to complete the whole process development. From breadboard prototype verification to the design of customized PCB expansion board, we finished hardware development, embedded firmware development, PC software development, and the whole process standardized testing and verification. The project completely covers four core learning outcomes of the module. All performance indicators reach the standard of household monitoring equipment. It can effectively fill monitoring gap in community and home rehabilitation scenes, providing a low-cost and high-reliability rehabilitation monitoring solution for patients, caregivers, and community medical staff.

1. Introduction and Research background

1.1 Context and Problem Statement

With the global aging population increased, the number of stroke and post-operative recovery patients has increased sharply. In the UK, stroke is a leading cause for disability. Although clinical rehabilitation is very important, because of the limitation of medical resources, most patients need to do long-term home rehabilitation after they leave hospital.

However, home rehabilitation lacking real-time monitoring has huge safety risks. Firstly, is the physiological load risk: patients without professional guidance can easily cause heart rate abnormality due to high exercise intensity, inducing secondary cardiovascular complications. Secondly is physical injury risk: because the coordination ability of patients is weak, the fall risk is very high. Once a fall happens and not rescued in time, it will cause serious physiological and psychological secondary damage.
With the promotion of NHS home rehabilitation services, about 83% of post-operative and chronic disease patients need to complete recovery in community or home. However, there are three core pain points in home rehabilitation:

1) **Rehabilitation monitoring resource gap**:
NHS community medical staff is not enough, so they cannot achieve high-frequency and regular monitoring of vital signs and progress. About half of home patients lead to longer recovery or re-admission because abnormal status is not found timely;

2) **Professional equipment is expensive and consumer-grade functions are not fit:**
Medical-grade monitors are high price and only fit for hospital. Consumer wearables only achieve basic life signs monitoring, without limb motion monitoring, fall warning and structured data records for rehabilitation. It cannot provide effective evaluation basis for medical staff;

3) **Rehabilitation compliance is hard to control:**
For monitoring means like heart rate and SpO2, medical staff cannot accurately evaluate recovery situation from long distance, leading to the rehabilitation plan cannot be adjusted dynamically.

The development of wearable medical technology provides a possible solution. Existing academic research has verified that based on ARM Cortex-M core microcontroller and mass-produced sensors, it can achieve

low-cost monitoring. Its accuracy can meet home measurement. This project is based on this research to develop a wearable auxiliary monitoring system to solve the core pain points.

1.2 Project Objectives

The project aims to develop a home rehabilitation monitoring based on STM32L476RG low-power microcontroller to achieve heart rate and SpO2 monitoring, temperature and humidity measurement, and achieve level meter function, two games, and gravity accelerometer. It provides a low-cost, high-reliability, and training monitoring solution for recovery patients.

2. Project Management

This is a two-person group project. We strictly follow PRINCE2 small project management framework, completely covering the whole life cycle: project starting, planning, executing, monitoring, and closing.

2.1 Team Division and Responsibility Allocation

This project is a two-person group. Members have clear division and specific task allocation to guarantee the progress:

Team member	Key role	Key responsibility	Key deliverables
Lowlife555	Head of Software and Algorithms	Embedded firmware development, implementation of core algorithms	Embedded firmware, algorithm simulation and implementation code, PC-based software
Mo Yan	Head of Hardware	Schematic and PCB design, hardware testing, project report	Hardware schematics, PCB layouts, BOMs, reports

Table 1 Team Roles and Responsibilities Table

2.2 Project Schedule

The development cycle of this project lasts from October 2025 to April 2026, advancing strictly according to the milestone nodes under the PRINCE2 framework. The table below lists the key tasks and estimated time of each stage in detail:

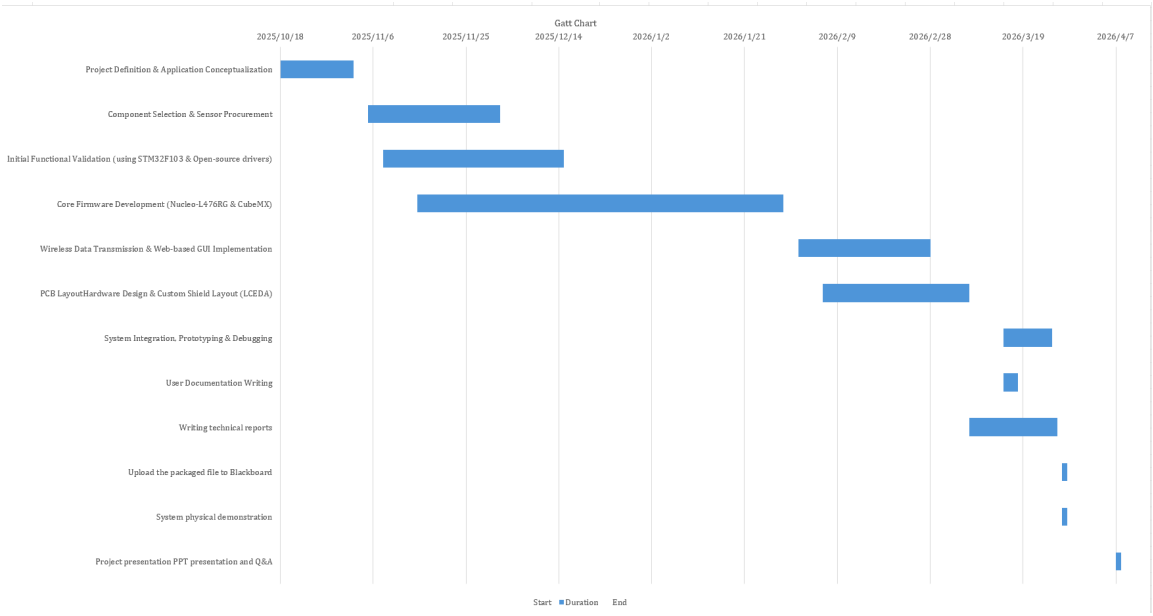


Figure 1 Gantt chart

3. System Core Design & Implementation

3.1 Hardware System and PCB Design

3.1.1 Overall Hardware Architecture

The system core hardware is STM32L476RG Nucleo development board, it meets the course requirement [only use STM32L476RG as main control] requirement, adopting modular hierarchical design (designed a PCB expansion board to integrate all the required sensors/display modules).

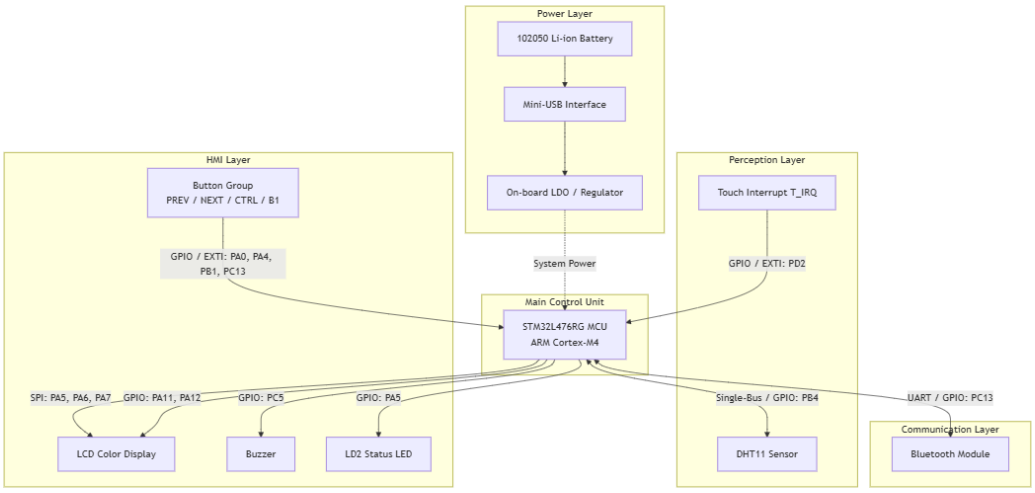


Figure 2 Hardware system overall architecture diagram

3.1.2 Selection of Core Components

All components selection is passed engineering argument, completely matching project needs and course requirements, the core selection is as shown in below table:

Component model	Core Functions	Selection Demonstration	Course compliance
STM32L476RG	Main controller	Course force specifies main control; Cortex-M4 core brings FPU, meeting sensor algorithm calculation needs; integrated SPI/I2C/UART/RTC etc. all required peripherals; low power feature fits portable scene.	100% meets the course force requirement, no other MCU use.
DHT11	Temperature and humidity monitoring	Common lab stock device; single-bus communication, occupy little IO; measurement range covers indoor home all scenes; calibrated digital output no need extra conditioning circuit.	Meets the course component usage requirement.
MAX30102	Heart rate and blood oxygen monitoring	Common lab PPG sensor; I2C interface, integrate LED driver and photodiode; special design for wearable health monitor, low power high sensitivity.	Meets the course component usage requirement.
MPU6050	Motion/Posture Monitoring	Common lab 6-axis IMU; I2C interface, integrate DMP motion processor; meet step counting, fall detection, posture calculation all function needs.	Meets the course component usage requirement.
ILI9341 TFT LCD	Human-computer interaction display	Common lab 2.4-inch TFT LCD; SPI interface, 240x320 resolution, 16-bit colour depth; big size, high colour display, fit elderly users vision needs and game interface rendering.	Meets the course component usage requirement.
XPT2046	Resistive touch control	Common lab resistive touch controller; SPI interface, share SPI bus with ILI9341; support touch coordinate collection, realize buttonless menu navigation.	Meets the course component usage requirement.
HC-05	Bluetooth wireless communication	Common lab Bluetooth module; UART interface, 9600 baud rate stable transmission; support slave mode, fit PC side web GUI data interaction.	Meets the course component usage requirement.

Table 2 System hardware selection and course compliance analysis table

3.1.3 Circuit diagram design

The system circuit adopts sub-module design, all circuits are passed breadboard prototype verification before doing schematic design, finally solidified to PCB.

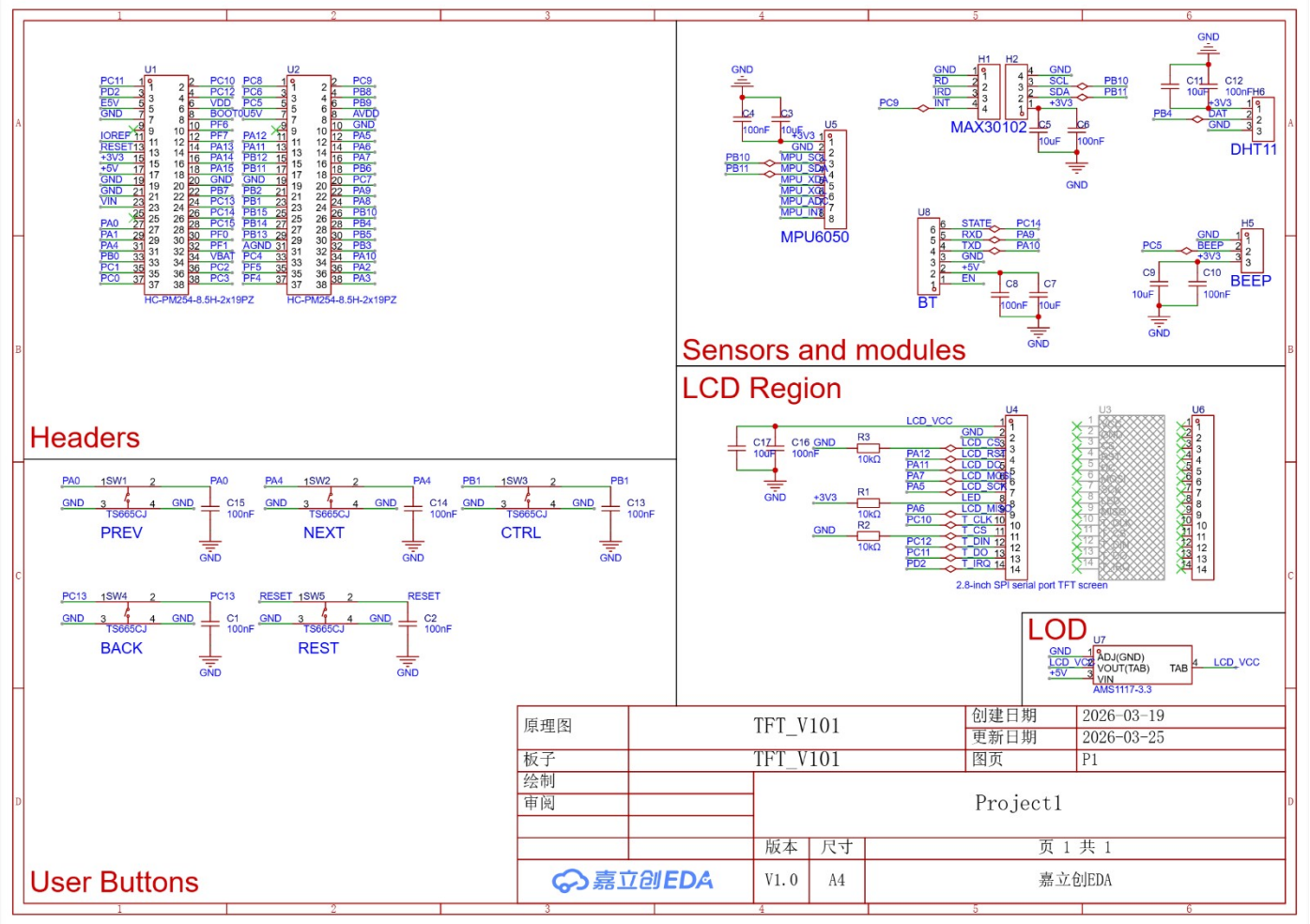


Figure 3 Schematic design diagram

Core Circuit Design Key Points

1) Use Net Label to show the link relationship between pins, avoiding messy wiring waste schematic space.

2) Power Management and Voltage Conversion
   a) **Low Dropout Regulator:** The system adopts AMS1117-3.3 regulator chip (U7), converting 5V input to stable 3.3V. It provides stable 3.3V power supply for the LCD screen.

   b) **Decoupling and Filtering:** Our team implemented 100nF and 10uF capacitor decoupling networks at the MCU interface, sensor modules, Bluetooth module, and LDO input/output ends. This combination can filter high-frequency switching noise and smooth low-frequency power ripple at the same time, guarantee the stability of I2C and SPI high-speed communication, and promote the signal-to-noise ratio of signal acquisition.

3) Sensor Architecture
   a) **I2C Bus Multiplexing:** MAX30102 heart rate oxygen sensor and MPU6050 six-axis gravity acceleration sensor share one I2C bus of STM32L476RG (PB10/SCL, PB11/SDA). Adopting a

master-slave communication architecture, it large-scale simplifies hardware wiring and GPIO occupy.

b) **Single-bus Interface:** DHT11 temperature and humidity sensor connects to PB4. The circuit is match with C11/C12 filtering to ensure the anti-interference ability of data acquisition.

4) Display and Human-Machine Interaction

a) **SPI TFT Interface:** Adopting 2.8-inch SPI serial TFT display screen (ILI9341), finishing picture data transmission with microcontroller through SPI1 bus.

b) **Signal Integrity Design:** A 10kΩ current-limiting resistor network is designed on the LCD data/command selection lines. It plays a port current-limiting protection role, and also can improve the quality of rising and falling edges of control signals, suppressing signal ringing.

5) Active Alarm Circuit

The buzzer driving circuit links at PC5 pin. Combining with the MPU6050_Detect_Fall function in the code, it can send out alarm when detecting user fall down.

3.1.4 PCB Design and Engineering Implementation

The hardware carrier of this system (PCB expansion board) has experienced three major version iterations from prototype verification to finished product. The design process follows IPC-2221 design standard, on the core premise of ensuring electrical performance, signal integrity, and long-term operation reliability, fitting the target users' human-machine interaction needs and equipment structure assembly constraints.

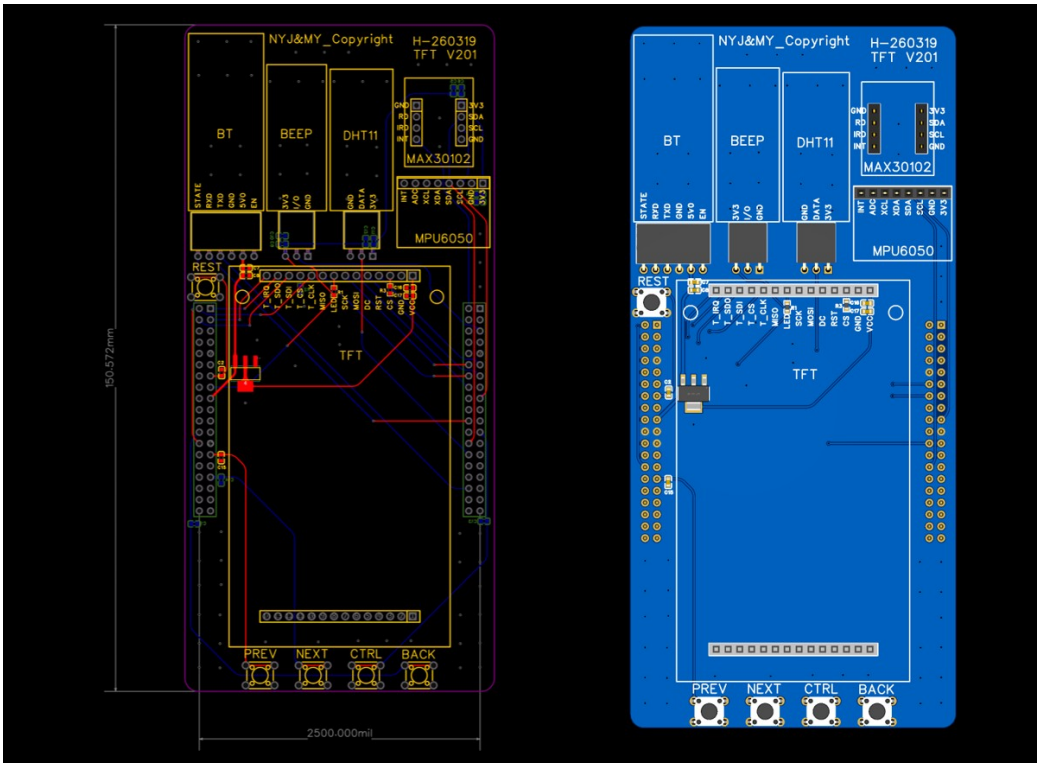


Figure 4 PCB expansion board design diagram

1) PCB Layout Design Core Points

PCB layout stage takes functional modular partitioning, signal link shortest, heat and electromagnetic isolation, and ergonomics adaptation as four core design principles, while at the same time strictly matching the 2*19pin header distance (2500mil) of Nucleo-MB1136 development board.

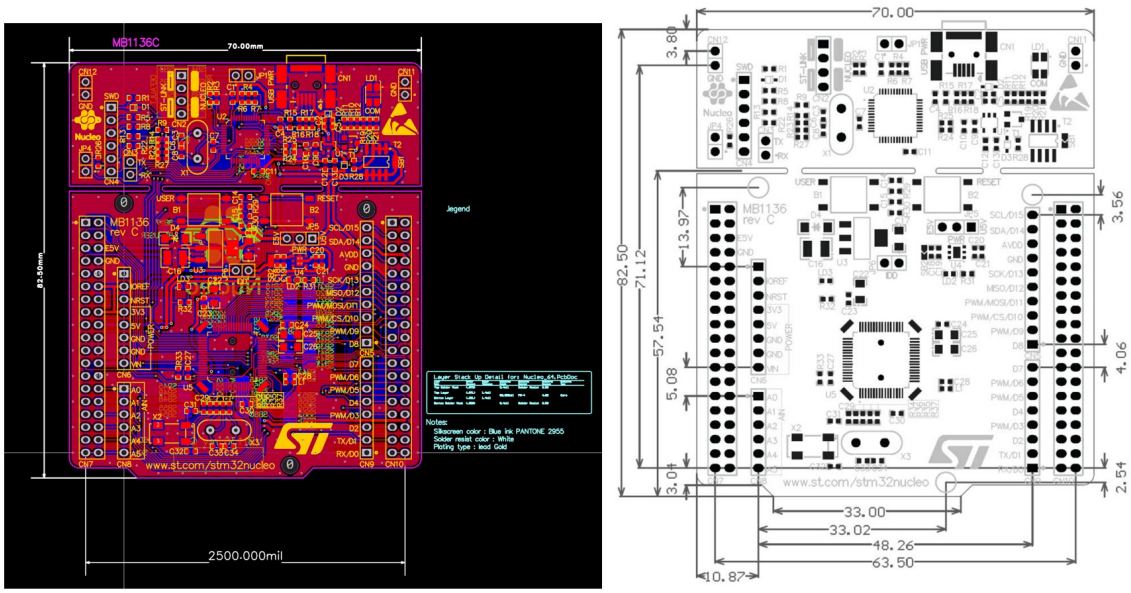


Figure 5 User manual and MB1136 system board downloaded from the ST website

2) Functional Modular Partitioning Layout Design

a) Development board interface area:

The Nucleo pin header pads on the left and right sides of the PCB strictly match the original 2.54mm pitch size; all peripheral lead-out traces are directly led out from the corresponding pads, no extra winding, ensuring signal link shortest.

b) Sensor functional area:

Concentrated layout in the PCB top area, MAX30102 heart rate oxygen sensor and MPU6050 six-axis gravity acceleration sensor are placed close to the main control I2C bus pins. The core design purpose is to shorten the I2C bus physical length, increasing the timing synchronisation in the scene of multiple sensors hanging on one bus.

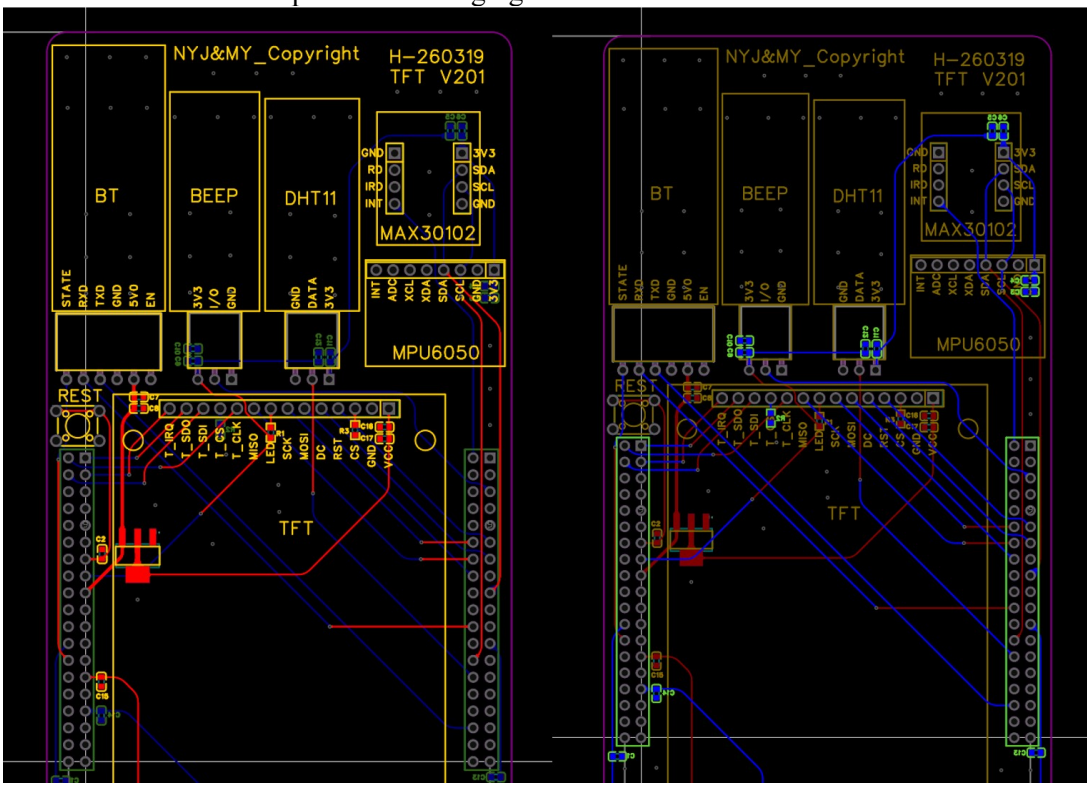


Figure 6 Top and Bottom view of PCB board

c) TFT display area:

Layout in PCB centred position, so both can let the designed finished product more look like a product, fitting user operation habit of mobile phones, tablets and other smart devices, rather than a rough "DEMO". On the technical level, this design also can guarantee screen ribbon cable no extra bending. Utilizing the gap between ILI9341 and PCB after plugging into female header to place AMS1117-3.3 regulator chip.

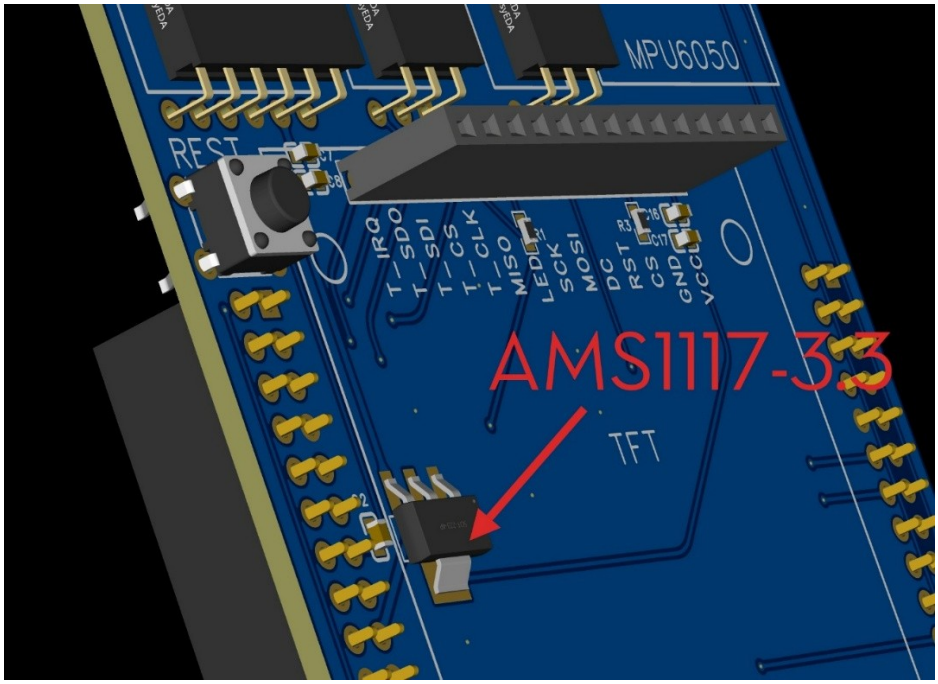


Figure 7 3D preview of ILI9341 region

d) Using Curved Traces in PCB design:

Traces all adopt smooth circular arc traces (Curved Traces), compared to traditional 45° or 90° broken line traces (Sharp Corners), it has significant advantages in signal integrity (Johnson & Graham, 1993), electromagnetic compatibility and manufacturing reliability (IPC, 2012).

e) User interaction button area:

Layout common user buttons at the PCB bottom edge, can fit users' natural holding posture when hand-holding this device, lowering learning cost. Users' not common RESET button is placed near the main control interface area, convenient for maintenance personnel to fast reset system when debugging/calibrating sensors.

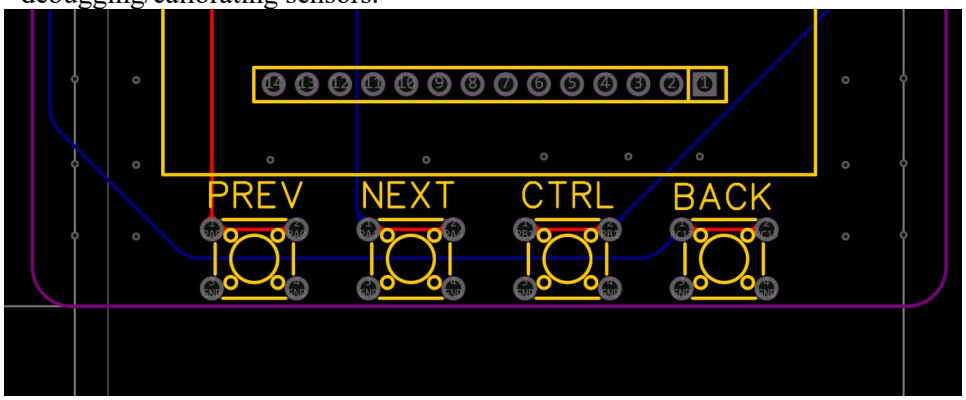


Figure 8 User interaction button area

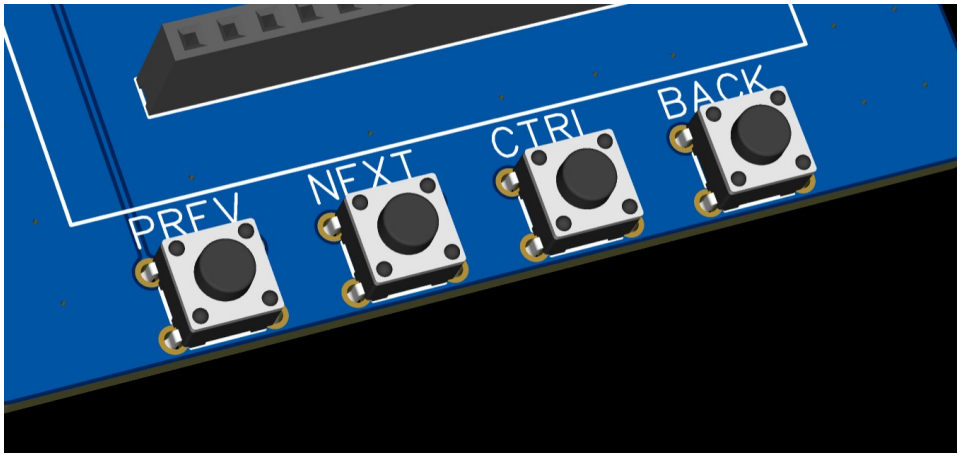


Figure 9 3D preview of the user interaction button area

3) PCB Development and Iteration Process:

The PCB design of this project has experienced three versions of design iteration, reserved sufficient time to do [Requirement Verification - Trial and Error Optimization - Engineering Implementation] process, reflecting the design thinking change and thinking of user needs during the learning and development process.

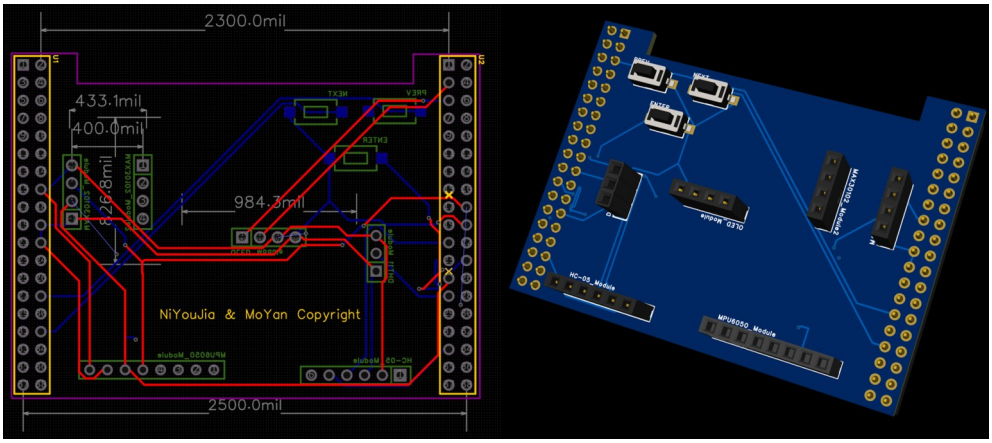


Figure 10 The First version PCB

a) In the initial design:

Our team did not spend much time on polishing details and module layout. The core goal was to provide a more convenient hardware foundation for developers after confirming module selection, replacing DuPont wires to connect various components. During debugging, DuPont wires were quite loose and easy to fall off during movement, and there was no need to waste time on reconnecting.

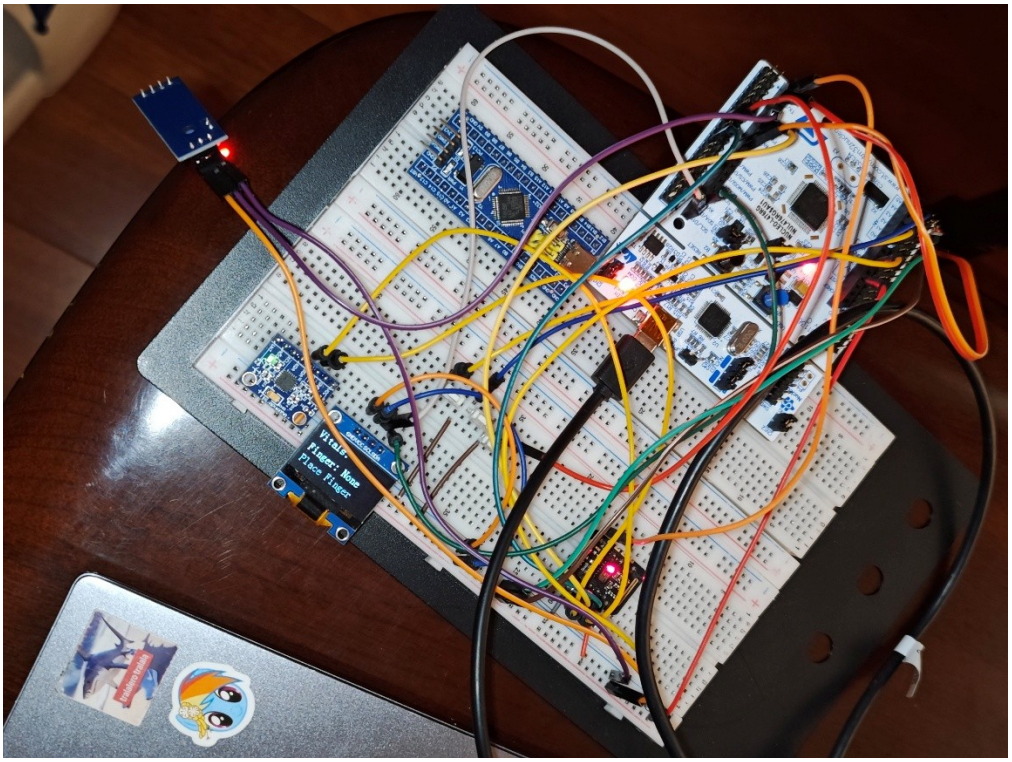


Figure 11 Use DuPont wires to connect the various modules.

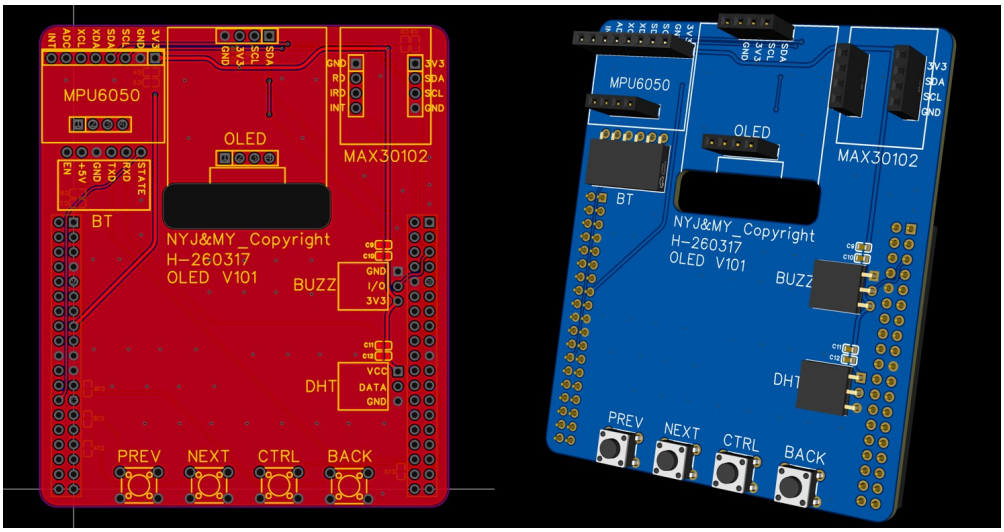


Figure 12 Second version PCB

b) In the second version of the PCB design:
Our team considered more aesthetic and operational convenience, and added more details, for example: adding pin names in the silkscreen layer, adding decoupling filtering structures.

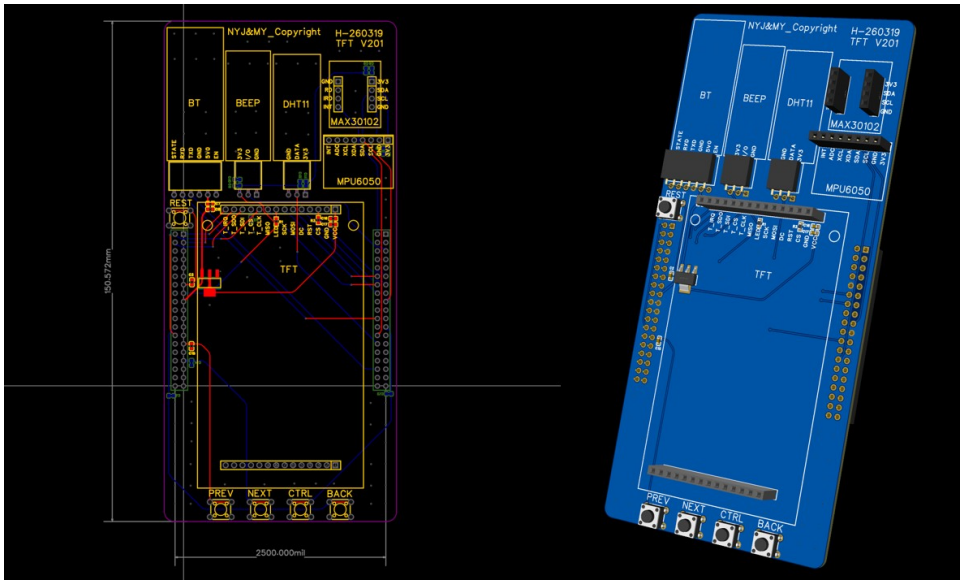


Figure 13 Third version PCB (LCD Screen)

c) Final version

Combined with our design intention and user needs, we all agreed that adopting a touch colour screen in the final version is better. Compared to the I2C 0.96-inch OLED screen, it has a larger display area and higher colour resolution (320*240). This design change not only can clearly present dynamic heart rate and blood oxygen data, but also provides a larger physical visual space for system built-in rehabilitation auxiliary games (such as Maze Ball, Dino Jump), greatly improving user experience.

4) Engineering testing phase:

However, in the engineering test stage, we faced a severe Engineering Trade-off. Limited by the SPI bus transmission rate of STM32L476RG, the huge data amount of 2.8-inch LCD caused the screen refresh rate to be significantly lower than the early I2C OLED version. Especially when running Dino Jump and Maze Ball which need high-frequency redrawing, the lower frame rate brings certain visual delay.

After our many times consideration and evaluation, the final scheme of adopting LCD colour screen was finally established. The engineering decision logic is as following:

a) User-Centric Design:

The core audience of this device is community inpatients and elderly people who stay at home for rehabilitation. Vision degradation is a universal symptom of this group. The 0.96-inch OLED in previous version is very small, which may cause serious reading obstacles. In the core requirements of medical auxiliary devices, the priority of interface clarity, high contrast and big font display is absolutely higher than higher dynamic refresh rate.

b) Core Function Adaptation:

The primary function of the system is basic life signs monitoring and environment data monitoring. The display of these data does not need very high refresh rate, and LCD is enough to perfectly complete the task.

c) Software Algorithm Compensation:

To release the SPI bandwidth bottleneck of hardware level, I have carried out deep optimization on software architecture. Inspired by the compression standard of H.264 video coding, I used an incremental update strategy in the driver of this system. By the hardware window configuration "LCD_Set_Window", only the pixel areas where coordinates have changed are erased and redrawn. This realizes a local refresh mechanism, reduces the redundant load of SPI bus, and achieves a higher dynamic refresh frame rate under limited bandwidth.

4. Software Architecture & Algorithms

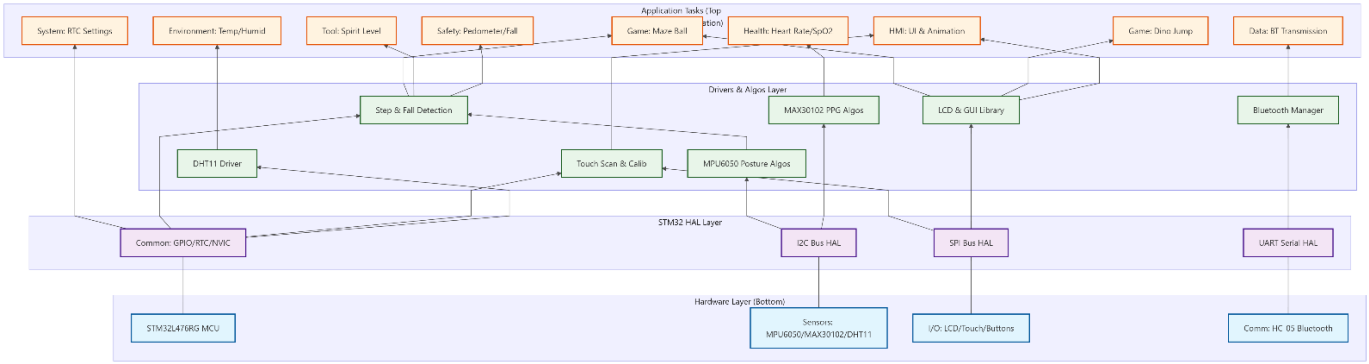


Figure 14 Diagram of the overall software architecture

4.1 Overall Software Architecture

This system adopts a bottom-to-up 4-layer modular architecture. Such architecture is easy for independent debugging, function expansion and problem locating of each sensor and functional modules. The architecture levels are as following:

1) Hardware Abstraction Layer:
Based on STM32 HLA library to realize bottom drivers of on-chip peripherals, including I2C, SPI, UART, GPIO, RTC, DWT hardware timers, etc. It provides unified hardware operation interface for upper code drivers and ignored the differences of bottom registers.

2) Peripheral Driver Layer:
The core layer of system bottom, containing independent drivers of three sensors, LCD screen, touch, and Bluetooth module. For sensor drivers, this layer completed functions like sensor initialization, original data reading, and basic data verification, providing stable and reliable data source for upper algorithms and applications.

3) Algorithm and Data Processing Layer:
Realize core data analysis and algorithm calculation based on sensor original data. This is the core of function landing. The related algorithms of MPU6050 and MAX30102 modules take the core development amount of this layer, including posture angle calculation, step counting algorithm, digital filtering, and heart rate/SpO2 calculation.

4) Application and Task Scheduling Layer:
Top logic layer, including main loop time-slice polling scheduling, UI interface refreshing, menu interaction, application entry management, Bluetooth data transmission, and button/touch event response. This layer set independent menu entries for core functions of device, realizing function switching, status management and human-machine interaction.

The system main loop adopts time-slice polling scheduling mechanism. According to real-time requirements of functions, it allocates independent running cycles for different modules. The MPU6050 data reading cycle is 30ms, firstly protecting the real-time of step counting and fall detection, and the operability of motion games. The reading cycle of MAX30102 is 8ms, DHT11 sampling cycle is 2000ms, and UI refreshing cycle is 30-100ms, avoiding low priority tasks blocking high priority functions.

4.2 Debugging and Function Implementation of Core Sensor Modules

4.2.1 MPU6050 Six-Axis Attitude Sensor

MPU6050 is the core sensor responsible for rehabilitation movement monitoring, safety early warning and interactive rehabilitation training in this system. It is the key component to realize balance ability evaluation, step counting, fall abnormal warning, and interactive game (Maze Ball).

Core Principles:

1) Communication Principle:

I2C master-slave communication architecture. In this system, the 7-bit slave address is 0x68. The chip identity and communication link verification are completed through the WHO_AM_I register (address 0x75). The sensor and MAX30102 share the I2C2 bus of STM32L476RG to realize a one-master-multi-slave hardware architecture, which greatly reduced the GPIO occupation.

2) Range and Conversion Principle:

Combining with the measurement requirements of fall detection in actual scenarios, the system set the accelerometer range to $\pm 8\,g$, corresponding to the conversion coefficient of $4096\ \ LSB/g$. The gyroscope range is $\pm 250\,dps$, and the corresponding conversion coefficient is $131\ \ LSB/(°/s)$. The raw data is output through the chip's built-in 16-bit ADC, and then converted into standard physical quantities through formulas.

3) Core Algorithm Principles:

a) Attitude Angle Calculation:

Through the three-axis acceleration data, calculate Roll and Pitch angles to provide data for balance ability evaluation of level function and Maze Ball gravity rehabilitation game. The core formulas are as following:

$$Roll = -\arctan2(A_y, A_z) \times \frac{180}{\pi}$$

$$Pitch = \arctan2\left(-A_x, \sqrt{A_y^2 + A_z^2}\right) \times \frac{180}{\pi}$$

b) Step Counting Algorithm:

Adopt the four-step logic of "Resultant acceleration calculation - Low-pass filtering - Peak detection -Time anti-repetition". Through low-pass filtering to filter movement noise, setting 1.6G acceleration trigger threshold and 400ms minimum step interval to reduce data error.

c) Fall Detection Algorithm:

Based on the complex safety requirements of this sensor, set the impact determination rules: (acceleration value of any axis $>\ 3G$, or X/Y axis $>\ 2G$ at the same time). After detecting the fall features which meet the rules, the sound and light alarm will be triggered.

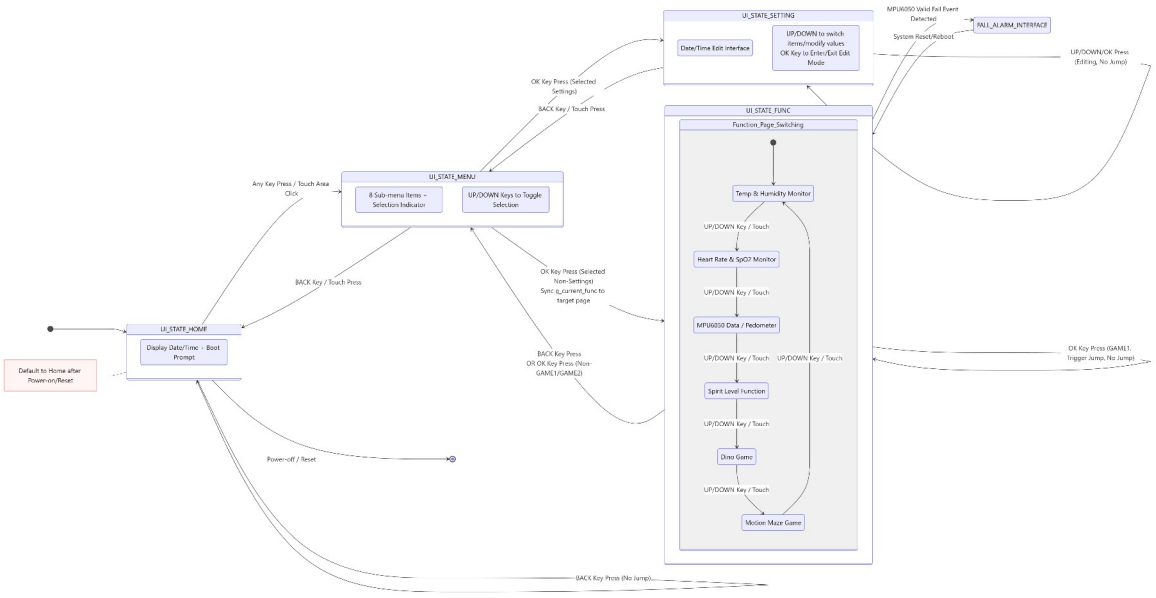


Figure 15 System UI Finite State Machine Transition Diagram

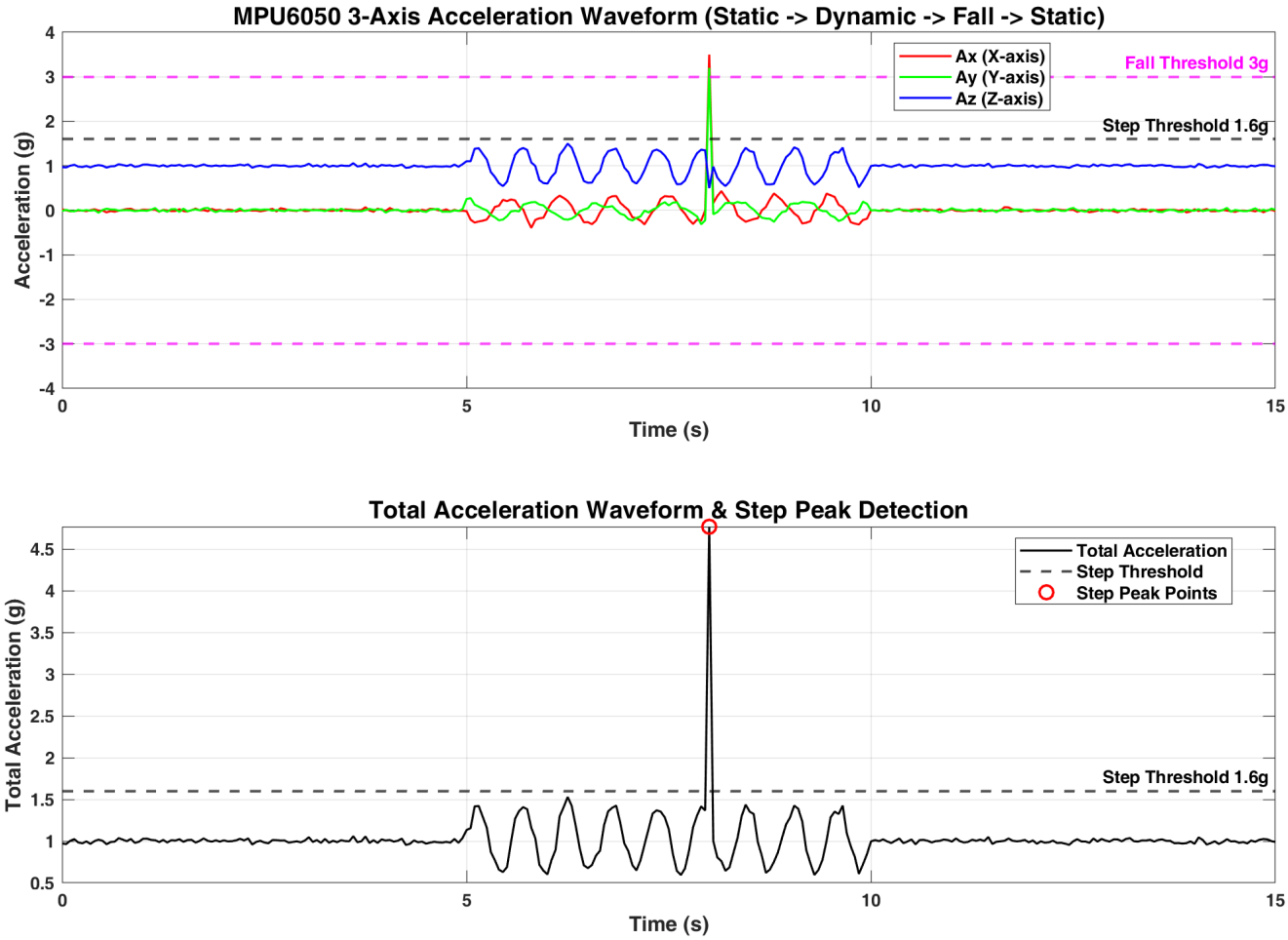


Figure 16 MATLAB plot of measured step count numerical waveforms

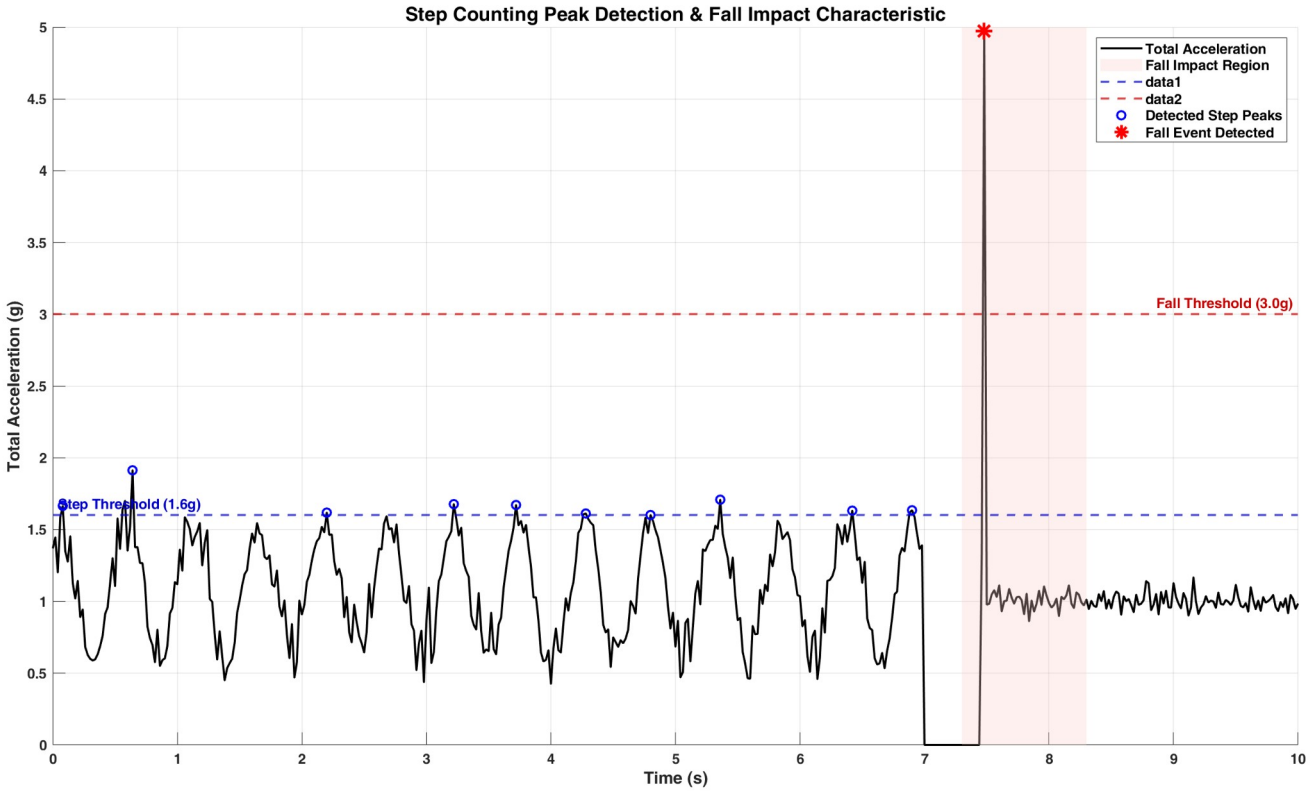


Figure 17 Measured characteristic waveforms of fall impact plotted using MATLAB

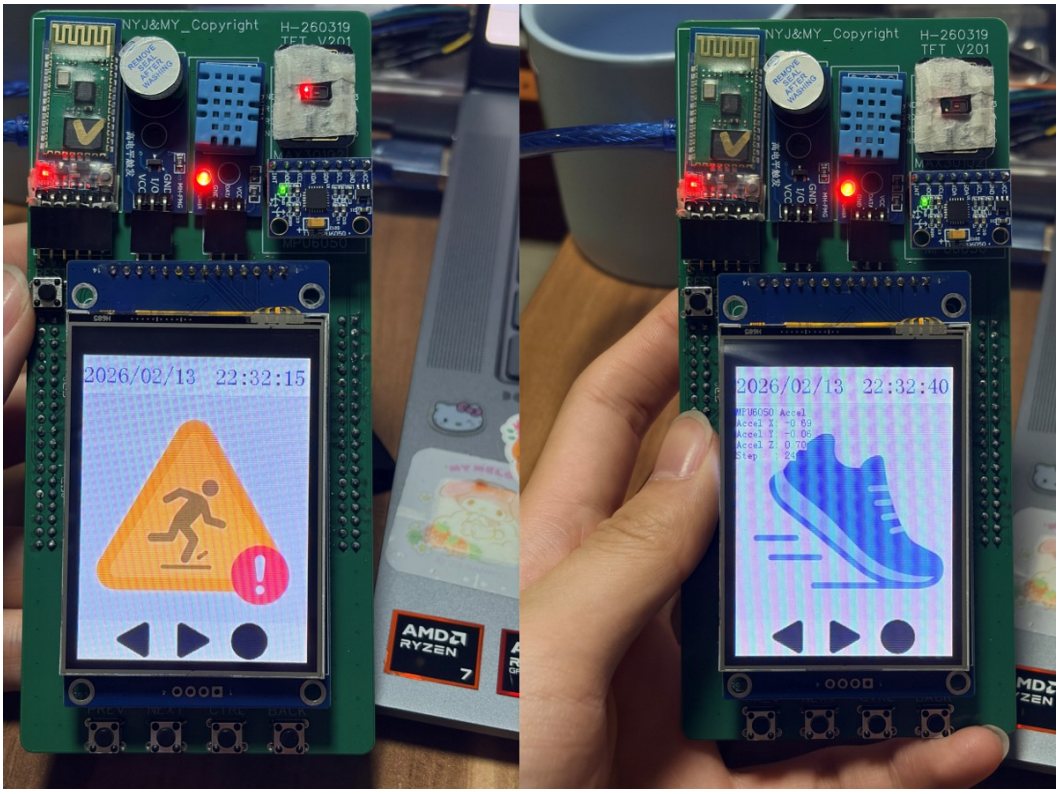


Figure 18 Real-world photos of the step counting and fall detection functions

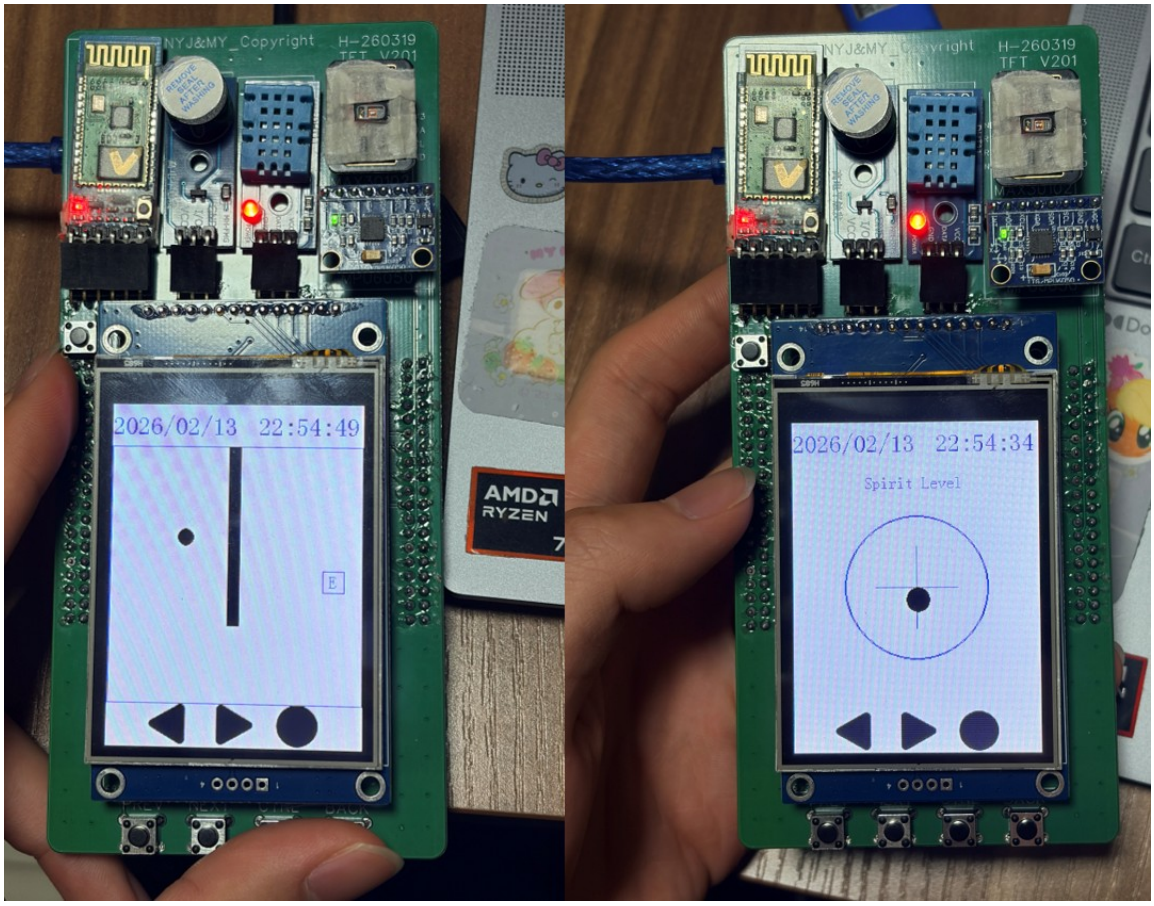


Figure 19 Maze Ball and Spirit Level functionality implemented using MPU6050

4.2.2 MAX30102 Heart Rate and SpO2 Sensor Debugging Process

To meet the life signs monitoring requirements of patients in rehabilitation periods, such as post-operation, chronic diseases and sports injuries, I use MAX30102 to realize the function of heart rate and SpO2 detection.

1) Hardware and Communication Basic Debugging:

This sensor and MPU6050 share the I2C2 bus of STM32L476RG, adopting a one-master-multi-slave communication architecture. The device address is 0xAE. In the debugging stage, first reading the PART_ID register (address 0xFF), and then completing the configuration of sensor registers. Setting SpO2 work mode, sampling rate 100Hz, and LED driver current about 7mA. Finally realize the stable reading of FIFO queue raw data, with sampling cycle 8ms, fitting the system time-slice polling scheduling mechanism.

2) Core Signal Processing and Algorithm Implementation:

a) DC Component Removal:

Adopting IIR filter ($\alpha = 0.95$) to extract DC baseline of IR and Red signals, separating the AC component containing pulse information, and eliminating the inherent interference such as environment light and static environment light.

b) Digital Noise Reduction Filtering:

Taking low-pass filtering ($\alpha = 0.7$) for AC component to filter high-frequency movement noise, improving the signal-to-noise ratio of pulse wave signal.

c) Heart Rate Calculation:

The data exceeding threshold is recorded as peak data. Setting a minimum period of 500ms to avoid repeated peaks leading to misjudgement. Calculating instantaneous heart rate through adjacent peak interval, and outputting stable value through 5-point moving average (removing extreme values and calculating the average of the remaining three values).

d) SpO2 Calculation:

Based on Lambert-Beer Law, calculating the light absorption ratio R-value through peak-to-peak value. Adopting the calibration formula to calculate SpO2:

$$SpO_2 = 104 - 17 \times R + Calibration\ \ offset$$

After 10-point moving average filtering, the value is outputted. The value is limited in the range of 70% - 100% to avoid outputting abnormal values.

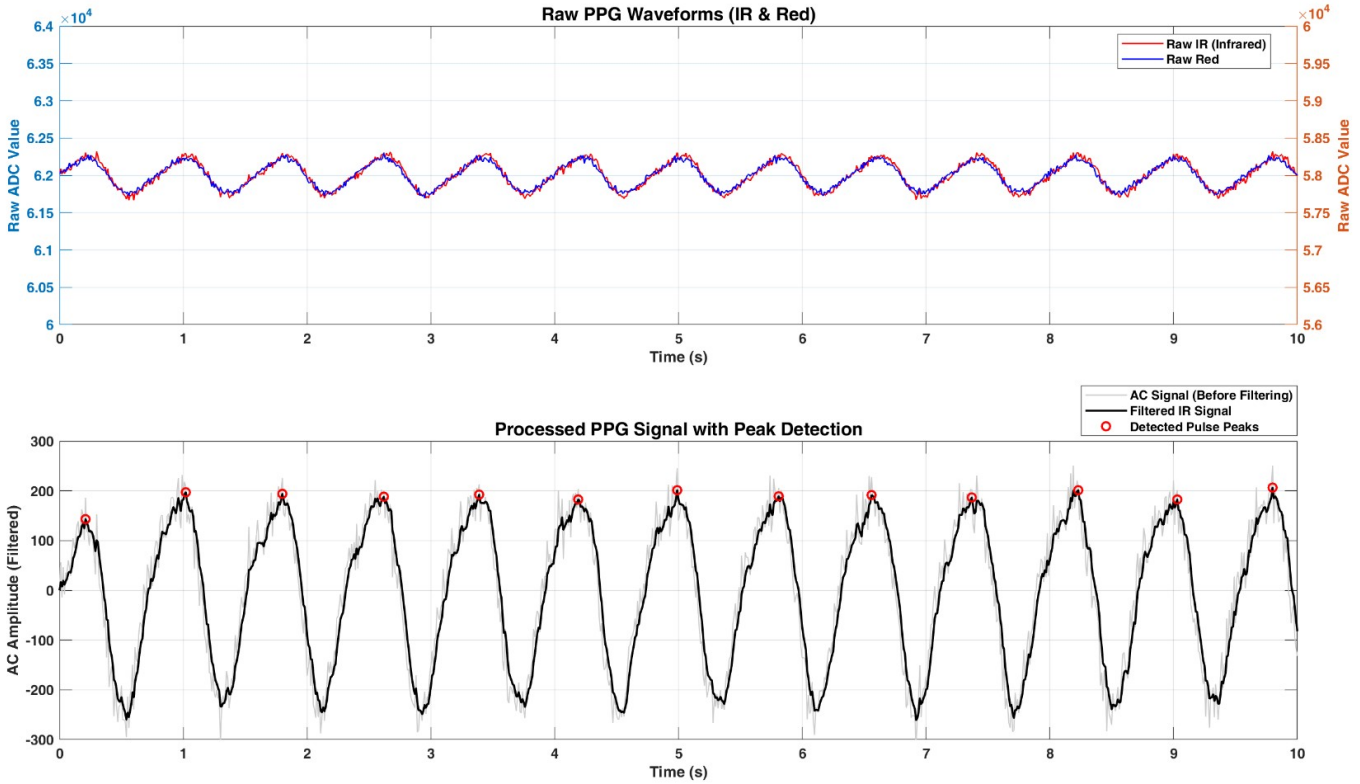


Figure 20 Use MATLAB to plot the waveforms of the data before and after PPG signal processing.

3) Debugging and Function Verification

After repeated testing and compensation values modification, under the condition of moderate pressure and static stable measurement, the heart rate error can be low to $\pm 5\,bpm$, and the SpO2 measurement error can be low to $\pm 2\%$ (comparing to the data measured by commercial smart watch Samsung Galaxy Watch4 Classic, see the figure below); It has already possessed reference value, and can assist users to judge body health status.

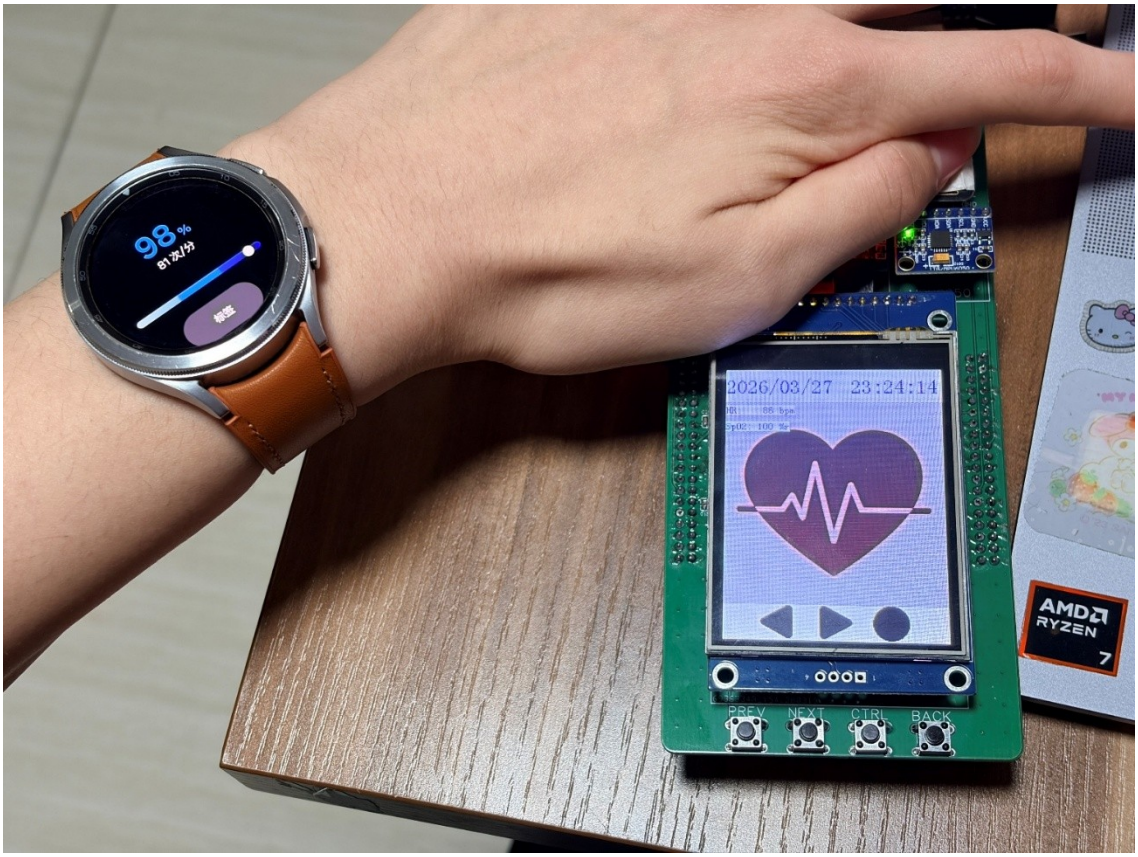


Figure 21 Comparison of actual photos of the features in action with data from smartwatches available on the market

4.2.3 Key points for debugging other sensor modules

1) DHT11:

This sensor adopts single-bus bidirectional communication. The development core is the dynamic switching of GPIO input/output mode and timing control. The start signal needs to output 19ms low level + $20\,us$ high level to avoid communication failure caused by timing deviation. After finishing 40bit data reading, it needs to verify and eliminate error code data. Setting 2s sampling interval to avoid the module no response caused by frequent reading. In debugging, checking the pin pull-up/down configuration to prevent data jump caused by level abnormality.

2) HC-05:

This module is based on STM32 USART1 serial port to realize wireless communication, with serial port parameters matching and standardised data frame design.

3) ILI9341 LCD Screen:

a) Hardware and Initialisation:

Data transmission is through SPI bus. Using the HAL_SPI_Transmit interface of HAL library to send commands and register values.

b) Screen Orientation and Display Memory Management:

By modifying the corresponding bits (Bit5-Bit7) of 0x36 register, the rotation of four directions (0°, 90°, 180° and 270°) can be realized. Meanwhile, the width and height in the global structure lcddev must be updated synchronously. Before drawing graphics or filling colours, setting the start and end coordinates of the drawing window (LCD_Set_Window) through 0x2A and 0x2B registers, then sending 0x2C command to prepare for writing into GRAM.

c) **Resistive Touch Panel (RTP) Calibration:**
Sending 0xD0 and 0x90 commands to the touch IC to read AD sampling values of X and Y respectively. To improve accuracy, the code adopted a filtering algorithm of multiple sampling and removing extreme values to take average. In the first time of use, performing four-point cross calibration. By calculating the scale factors (xfac, yfac) and offsets (xoff, yoff) between screen physical coordinate differences and AD sampling value differences, the touch panel AD value is accurately mapped to screen pixel coordinates.

5. PC Host GUI and Data Logging Design and Implementation

Based on the HC-05 Bluetooth module, this part developed a Python PC host software to realize wireless data transmission of the wearable rehabilitation monitoring system, and it included data export function. It provides traceable rehabilitation data support for home-based rehabilitators, caregivers and community medical staff.

Core Function Design:

Aiming at the home rehabilitation monitoring scenario, this PC host designed four core functions:

1) **Stable Bluetooth Communication:**
Establishing connection with the HC-05 module based on RFCOMM protocol, and the serial port parameters are matched with the lower computer.

2) **Real-time Visualization of Multiple Parameters:**
The PC host program can real-time display six core parameters: heart rate, SpO2, environment temperature and humidity, accumulated steps, and game scores.

3) **Real-time Information Display:**
Built-in log text box displays physical quantities after parsing raw data, which can be used for abnormality troubleshooting.

4) **Data Recording and Management:**
Setting an automatic saving mechanism with a 5-second cycle to prevent accidental loss. It supports manually exporting complete logs as .txt files. The log information contains timestamps, which is convenient for data archiving.

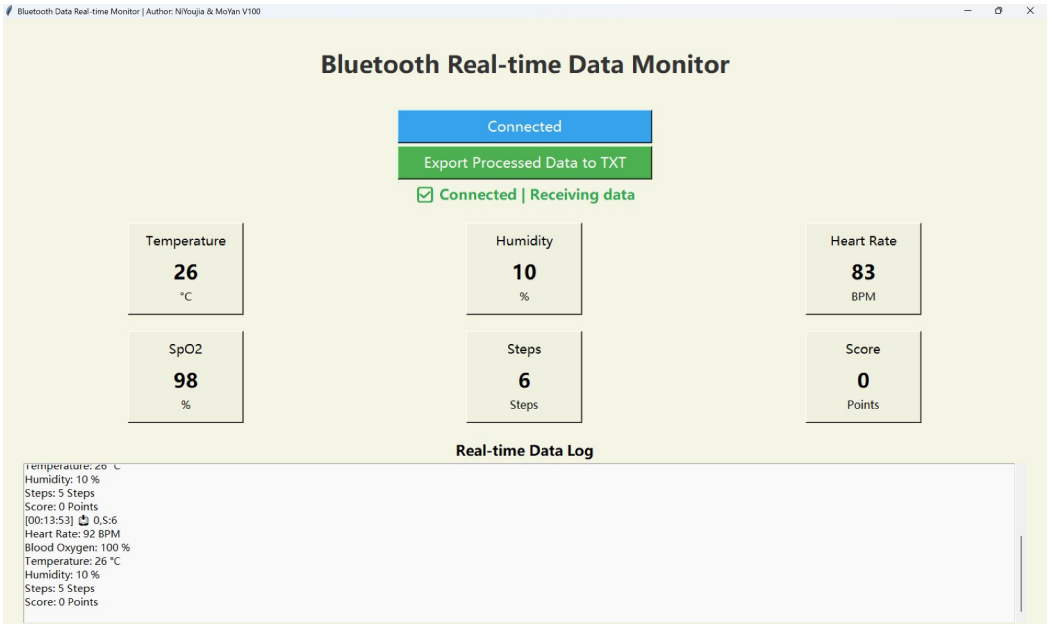


Figure 22 Host computer interface

```
[2026-03-28 01:13:30] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:2  Score:0
[2026-03-28 01:13:50] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:2  Score:0
[2026-03-28 01:14:10] Temperature:26  Humidity:10  HeartRate:66  SpO2:96  Steps:2  Score:0
[2026-03-28 01:14:30] Temperature:26  Humidity:10  HeartRate:65  SpO2:94  Steps:4  Score:0
[2026-03-28 01:14:50] Temperature:26  Humidity:10  HeartRate:66  SpO2:85  Steps:4  Score:0
[2026-03-28 01:15:10] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:4  Score:0
[2026-03-28 01:15:30] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:5  Score:0
[2026-03-28 01:15:50] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:9  Score:0
[2026-03-28 01:16:10] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:12  Score:0
[2026-03-28 01:16:30] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:12  Score:0
[2026-03-28 01:16:50] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:13  Score:0
[2026-03-28 01:17:10] Temperature:26  Humidity:10  HeartRate:0  SpO2:0  Steps:13  Score:0
```

Figure 23 Exported data

6. Intelligent Health Monitoring System Enclosure Design

Aiming at the landing use requirements of the intelligent health system, this project completed the full parametric 3D structural design of the device enclosure. It solved the core pain points that the prototype hardware with only PCB board has almost no protection and poor portability.

The design adopts a top-and-bottom split structure. The main body is divided into two modules: base cavity and top cover panel. After measuring data, the manufactured base cavity can adapt to the physical size of STM32L476RG core board and custom PCB expansion board, reserving enough space for battery power supply. The top cover panel made precise openings for TFT display screen, heart rate sensor and functional buttons, fitting the one-hand operation habit and adapting to the use needs of elderly and rehabilitation users.

PLA environmental protection material was selected. The enclosure edges carried out circular arc fillet optimization treatment, considering the structural protection, grip feeling and assembly convenience. It realized the evolution from prototype machine to a handheld device that can be landed for actual use.

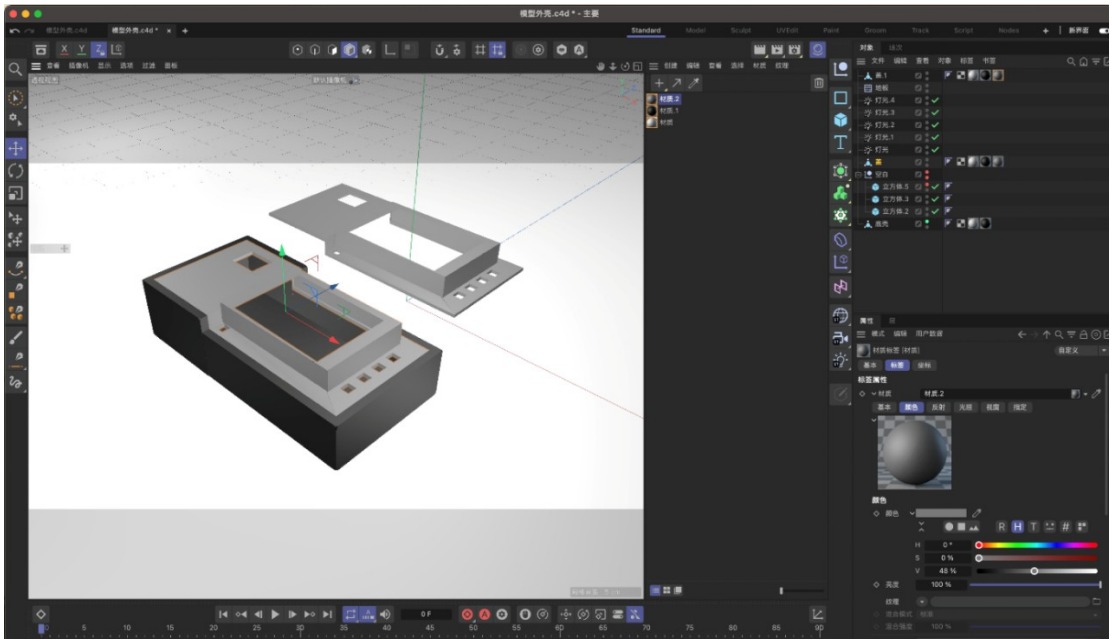


Figure 24 3D model of the shell design

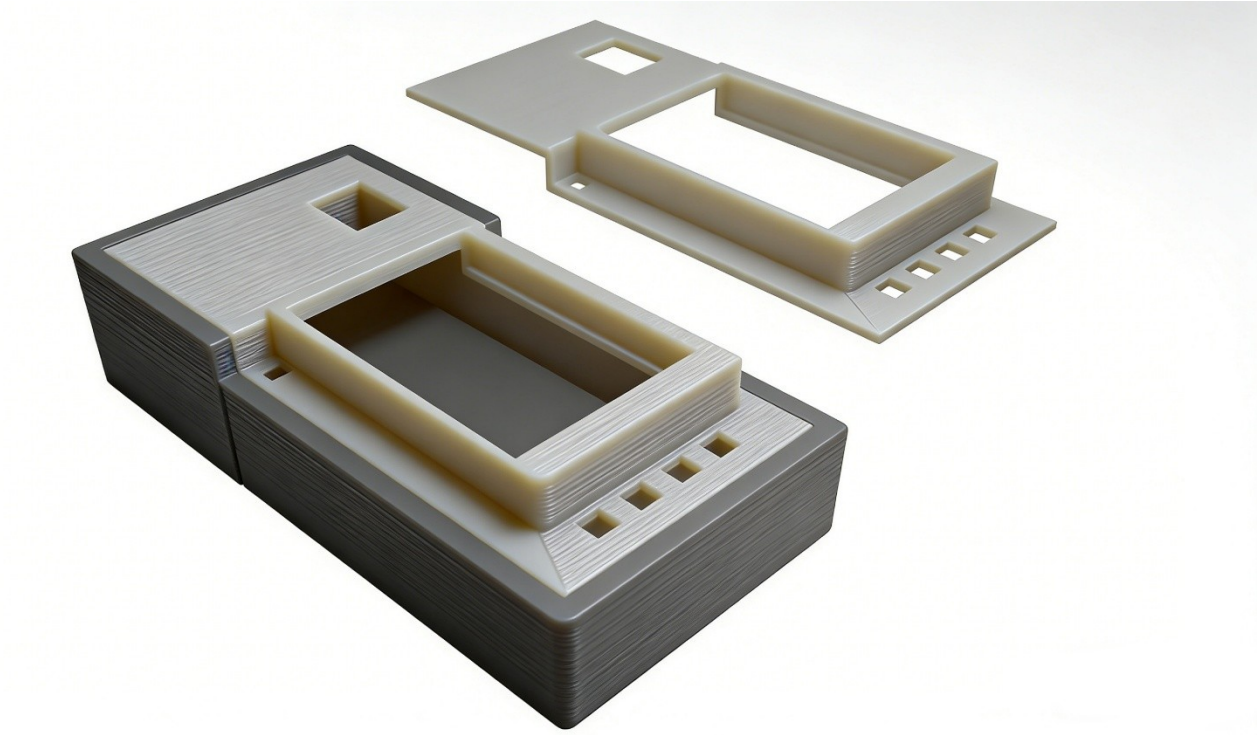


Figure 25 Reference image of the finished casing design (the finished product may differ).

7. One More Thing: Fully Customized Integrated Wristband Scheme

Aiming at the core pain points of the previous Nucleo MB1136 development board scheme, such as volume redundancy, no on-board power management and poor portability, we designed a more refined health monitoring system scheme as the engineering iteration direction of this project.

To fit the course requirements, this scheme uses STM32L476RG as the main control. Abandoning the development board architecture, we independently designed the main control minimum system, power management part, and Type-C power supply circuit. At the same time, it integrated all peripheral circuits of MAX30102 heart rate/SpO2, MPU6050 gravity acceleration sensor, and DHT11 temperature and humidity monitoring, integrating all functions into a single PCB.

Since independently designing all peripheral circuits can greatly improve integration and compress hardware volume, it realized the form upgrade from a handheld prototype to a wristband wearable device. Although it has not been successfully landed yet due to project period reasons, this design provides a theoretical basis for subsequent portability and wearable landing.

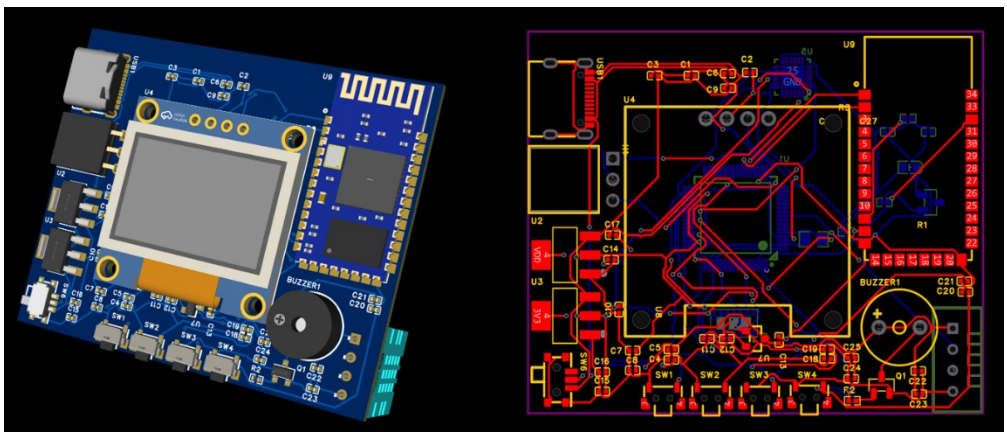


Figure 26 PCB design diagram for an integrated wristband solution

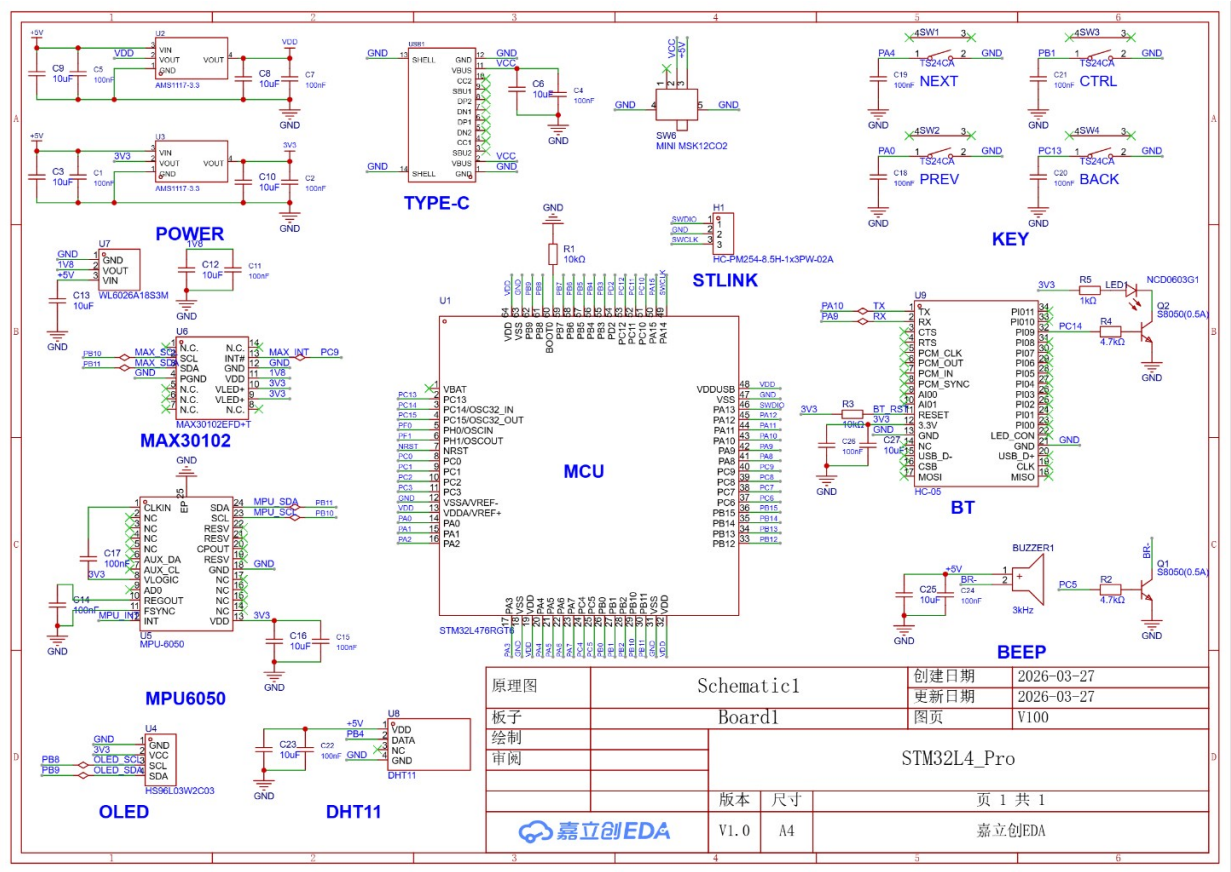


Figure 27 Schematic design for an integrated wristband solution

Smart Health Monitoring and Rehabilitation System ——WellSpan

User Manual

Compatible hardware: STM32L476RG core board & PCB expansion board
Target users: Inpatients in community hospitals, individuals undergoing home rehabilitation, and the elderly

University of Lancashire

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1. Product Overview

This intelligent health monitoring system focuses on the health data collection and the mental health protection. Through the multi-sensors, it achieves three core functions: physiological index monitoring, sporting ability evaluation, and interactive entertainment. The specific functions are as following:

- **Environment Monitoring:** Real-time collection of room temperature and humidity.
- **Health Monitoring:** Heart rate and SpO2 detection.
- **Motion Evaluation:** Gravity accelerometer data display/pedometer function, level meter function.
- **Entertainment Interaction:** Maze ball game, Google Dino game.
- **Safety Protection**: Fall detection.



Figure 1 Button diagram

2. Start-up and Initial Interface

2.1 Start-up Operation

After connect the device power, the system will start automatically. After waiting for a period of time (around 5s), it enters the home page.

2.2 Initial Interface Description

Information displayed on the home page:

- **Top:** Current date (form like: 20XX/XX/XX), time (form like: XX: XX: XX).
- **Bottom:** Prompt text "Press any Button to start".

3. Function operational Guide

3.1 Core operational Logic

- **Button Interaction:** Through "UP/DOWN Button" to switch menu options, "Confirm Button (CTRL)" enters the selected function, and "Return Button (BACK)" exits the current function or returns back to the previous level;

- **Status Switching:** All function pages can return to the menu page through "Return Button", and the menu page can return to home page through "Return Button";

- **Refresh Mechanism:** The system automatically refreshes data (frequency 1 time/second), no need for manual operation.

3.2 Accessing Function Menu

Press any button on the home page, the system enters the menu page. The menu includes 7 core function options. Switch the selected item (green circle mark) through "NEXT/PREV Button", and use "CTRL Button" to enter the corresponding function:

Menu option	Function Description	Corresponding sensor / module
DHT11	Temperature and humidity monitoring	DHT11 Temperature and Humidity Sensor
HR SpO2	Heart rate and blood oxygen monitoring	MAX30102 biosensor
MPU6050	Accelerometer / Step counting	MPU6050 six-axis sensor
Spirit Level	Spirit level (assessment of balance)	MPU6050 six-axis sensor
Dino Game	Google's T-Rex game	Buttons + LCD display
Maze Ball	Maze Ball Game (Gravity Control)	MPU6050 + LCD display
Setting	System time and date settings	RTC Real-Time Clock

Table 1 Function list

3.3 Detailed Operation of Each Function

3.3.1 Temperature and Humidity Monitoring (DHT11)

- **Access Method**: Select "DHT11" on the menu page, then press CTRL Button.

- **Interface Display**:
  - **Background**: Temperature and humidity theme image.
  - **Core Data**: "Temp: XX.X °C" and "Humi: XX.X%".

- **Operation**: Data refreshes automatically. Press BACK Button to return to the menu page.

3.3.2 Heart Rate and SpO2 Detection (HR SpO2)

- **Access Method**: Select "HR SpO2" on the menu page and press CTRL Button.

- **Operation Steps**:
  - The interface initially prompts "Finger: None/Place Finger".
  - Lightly place the finger on the MAX30102 sensor probe (keep stable, no need to use force, avoid shaking as much as possible).
  - The system prompts "Measuring… Wait: XX s".
  - After the countdown, it displays "HR: XXX bpm" and "SpO2: XXX %".

- **Exit**: Press BACK Button to return to the menu page.

- **Notice**: Finger should overlap the sensor during measurement, and avoid strong light direct shining on the probe.

3.3.3 Acceleration/Pedometer Monitoring (MPU6050)

- **Access Method**: Select "MPU6050" on the menu page and press CTRL Button.

- **Interface Display**:
  - **Background**: Pedometer theme image.
  - **Core Data**: Accel X, Y, Z (acceleration data of XYZ axes) and Step: XXX (accumulated steps).

- **Operation**: When the device moves with the human body, the pedometer data accumulates automatically. Press BACK Button to return to the menu page.

- **Note:** When using for the first time, keep the device steady for 1 second (to calibrate the sensor).

3.3.4 Spirit Level

- **Access Method:** Select "Spirit Level" on the menu page and press CTRL Button.

- **Interface Display:**
  - **Background:** Spirit level theme interface, including an outer circle and crosshair.
  - **Core Element:** A bubble, which moves with the device tilting.

- **Balance Evaluation Logic:**
  - **Bubble in center:** Good sense of balance.
  - **Bubble offset:** Larger offset means weaker balance (offset direction corresponds to device tilt direction).

3.3.5 Dino Game

- **Access Method: Select "Dino Game" on the menu page and press CTRL Button.**

- **Game Rules:**
  - Initial interface displays "DINO JUMP Press Key".
  - Press CTRL button to start game, the dinosaur moves forward automatically, and cacti move from right to left.
  - Press confirm button to control dinosaur "jump" to avoid cacti.
  - After colliding with cactus, the game is over and displays "Score: XXX".
  - Press confirm button to restart after game over, or press BACK button to exit.

- **Operation:** CTRL Button = Jump; BACK Button = Exit.

- **Notice:** Game speed increases with score.

3.3.6 Maze Ball

- **Access Method:** Select "Maze Ball" on the menu page and press CTRL Button.

- **Game Rules:**
  - The interface displays a maze map. The ball's initial position is centered, and the target point is at the bottom right corner.
  - Tilt the handheld device (using MPU6050 gravity induction) to control ball moving towards the target.
  - After the ball reaches target point, game is win. Press confirm button to restart.

3.3.7 Fall Detection

- **Functional Logic:** The system background detects acceleration changes in real-time through MPU6050. When fall characteristics (sudden acceleration change) are detected, it triggers sound and light reminder.

- **Operation:** No need to enter manually; it runs automatically after power on.

- **Reminder Method:** LCD screen flashes "DANGER" and the buzzer sound at same time.

- **Reset:** Press any key to clear the reminder.

3.3.8 Physical Button Function

This product takes that medical staff may need to use gloves for operation or users are not used to using the touch screen, the physical buttons of this product can replace the touch screen to operate all contents. The button functions represent "Previous", "Next", and "Control" respectively, which follows the same logic as the operation panel below the touch screen.

4. System setting

4.1 Accessing the Setting Page

Select "Setting" on the menu page and press CTRL Button to enter the time/date setting interface.

4.2 Entering/Exiting Edit Mode

- **Enter Edit:** The setting page initially displays "OK Edit B1: Back". Press CTRL button to enter edit mode (the interface prompts "UP/DN: Adj OK: Save");

- **Exit Edit:** Press the return button to give up modifications before saving. After saving, it automatically exits edit mode.

4.3 Editing Item Description:

Edit Item	Adjustment range	Example
Year	00–99 (corresponding to 2000–2099)	26 → 2026 (Year)
Month	01~12	08 →August
Day	01~31	15 → 15 (Day)

Hour	00~23	14 →2 PM
Minutes	00~59	30 → 30 minutes
Second	00~59	45 → 45 seconds

Table 2 Settings page editable items description

5. Precautions

5.1 Operation Safety

- **About power supply:** When not using the device own power supply, please use 5V/1A DC power to avoid overload;
- **Sensor protection:** The device has built-in sensors like MAX30102/DHT11, avoid collision and touching water;
- **Game duration:** When elderly users use interactive game functions, the single duration is suggested ≤ 10 minutes to avoid fatigue.

5.2 Maintenance and Care

- **Cleaning:** Use dry cloth to wipe device surface, prohibit using alcohol or disinfectant to contact sensor probes;
- **Storage:** Please put in dry and normal temperature environment when idle, avoid direct sunlight;
- **Calibration:** Every half year should through "MPU6050" function page to observe acceleration values (X/Y/Z should close to 0/0/1 when static). If deviation is large, please contact technical staff for calibration.

6. Troubleshooting Common Problems

Problem description	Investigation methods
Temperature and humidity data are not displayed	1. Check the wiring of the DHT11 sensor 2. Restart the device to reinitialise the sensor
Heart rate/blood oxygen measurement results were unavailable	1. Ensure your finger is in contact with the sensor 2. Avoid exposing the probe to direct strong light 3. Check the power supply to the MAX30102
Step count data is not accumulated	1. Ensure that the MPU6050 sensor has been calibrated 2. The device must move with the body (minor movements will not be counted as steps)
Game unresponsive	1. Check if the buttons are stuck 2. Restart the device to free up memory
The time setting is invalid after restarting	1. Confirm by pressing the "OK" button to save 2. Check if the battery is working properly (if configured)
False positive fall detection	1. Avoid subjecting the device to violent shaking 2. Contact technician to adjust the fall detection threshold

Table 3 Common Problems Checklist

7. Bluetooth data detection real-time host computer

Matching Device: WellSpan Intelligent Health Monitoring System (STM32L476RG Master + HC-05 Bluetooth Module)

Software Program: btV4.exe

- **Function Overview:**
  This PC host is the matching data monitoring software for the WellSpan Intelligent Health Monitoring System. Through the HC-05 Bluetooth module, it establishes wireless communication with the lower computer device to realize real-time visualization, automatic archiving, and one-key export functions of the collected data, providing support for rehabilitation monitoring data traceability.

    - **Core Functions:**
      1) **Real-time synchronization of 6 core monitoring data**: ambient temperature, ambient humidity, heart rate, SpO2, accumulated steps, and game scores.

      2) Automatic background data recording with a 20-second cycle to prevent data loss caused by abnormal exit of the software.

      3) One-key export of the data after standardized processing, convenient for data archiving and secondary analysis of data.

    - **Preparation before Running:**
      1) **System Requirements:** Windows 10 and above systems.

      2) **Hardware Requirements:** Computer equipped with internal/external Bluetooth adapter.

      3) **Storage Space:** At least 100MB of available disk space

      4) **Bluetooth Pairing:** Before running the software, complete the pairing of the HC-05 module in the Windows system settings. The default pairing code is 0000 or 1234. Data connection can be established only after successful pairing.

- **Core Operation Guide**
    - **Software Running:** Save btV4.exe to the local computer, double-click to run directly.

    - **Device connection:**
      1) Power on the WellSpan system and confirm HC-05 works normally.

      2) After running the software, click the [Connect HC-05 & Read Data] button to initiate connection.

      3) The status bar shows green [Connected ｜ Receiving data] means the connection is successful. If connection fails, you can check the error prompt in the bottom log.

    - **Data Export:** After the monitoring is finished, click the [Export Processed Data to TXT] button on the interface. Select the saving path in the pop-up window. After clicking confirm, all data received this time can be exported as a TXT file.

    - Precautions:
      1) The monitoring data displayed by this software is only for rehabilitation auxiliary

reference and does not have medical diagnosis qualification.

2) Do not force close the software during data recording. It is suggested to exit after finishing data export.

3) Non-professionals please do not modify other parameters in the software except MAC address and channel number, to avoid abnormal software performance.

Medical Compliance Statement

This project is a medical auxiliary rehabilitation equipment, it must be added a complete disclaimer according to UK MHRA regulations. The device is only for rehabilitation auxiliary reference, and it does not replace professional medical diagnosis and treatment. If the report and user guide miss this part, it is a compliance defect. It will be judged as "not considering the social and compliance risks of engineering activities" when grading.

Version number： 20260325V1.3
Last updated： 3/25/2026
Technical Support：[redacted]；[redacted] (email address)

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appendix:

1.

new_20260324-TFT.zip

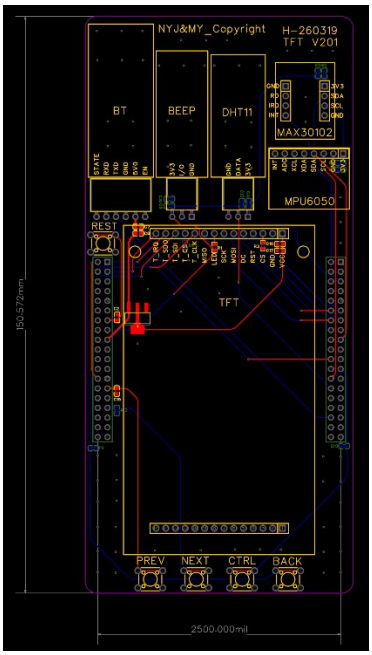
File 1 WellSpan_keil project files

2.

SCH_TFT_V101_2026-03-28.pdf

File 2 WellSpan Schematic PDF

3.



File 3 WellSpan PCB Image

4.

Gerber_PCB2_V101_2026-03-28.zip

File 4 WellSpan PCB Gerber File

5.

btV4.exe

File 5 Bluetooth host computer executable file

6.



btV4.py

File 6 Python-based host computer software (GUI) for real-time biometric data monitoring and archival

7.



max30102_ppg_
data.txt

File 7 Measured data

8.



plot_ppg_proces
sing.m

File 8 MATLAB script for PPG signal digital filtering and heart rate peak detection

9.



mpu6050_data.tx
t

File 9 Measured data

10.



plot_fall_detectio
n.m

File 10 MATLAB script for MPU6050 3-axis motion characterization

11.



plot_mpu6050_d
ata.m

File 11 MATLAB script for MPU6050 3-axis motion characterization

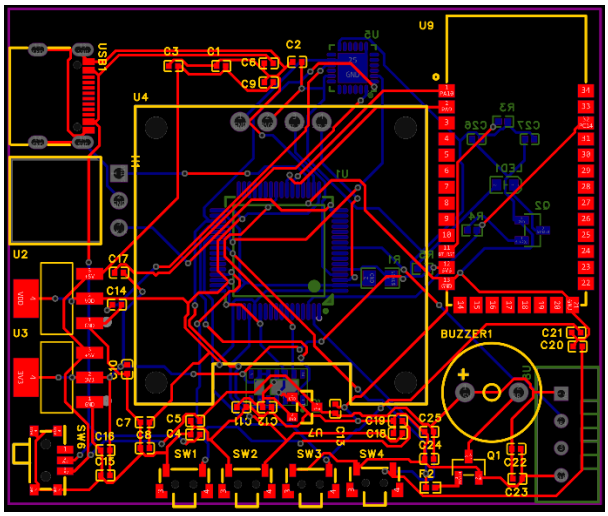
12.



SCH_Schematic1
_2026-03-28.pdf

File 12 One More Thing Schematic PDF

13.



File 13 One More Thing PCB Image